OVERVIEW OF THORAX

SKELETON, WALL, INLET, OUTLET

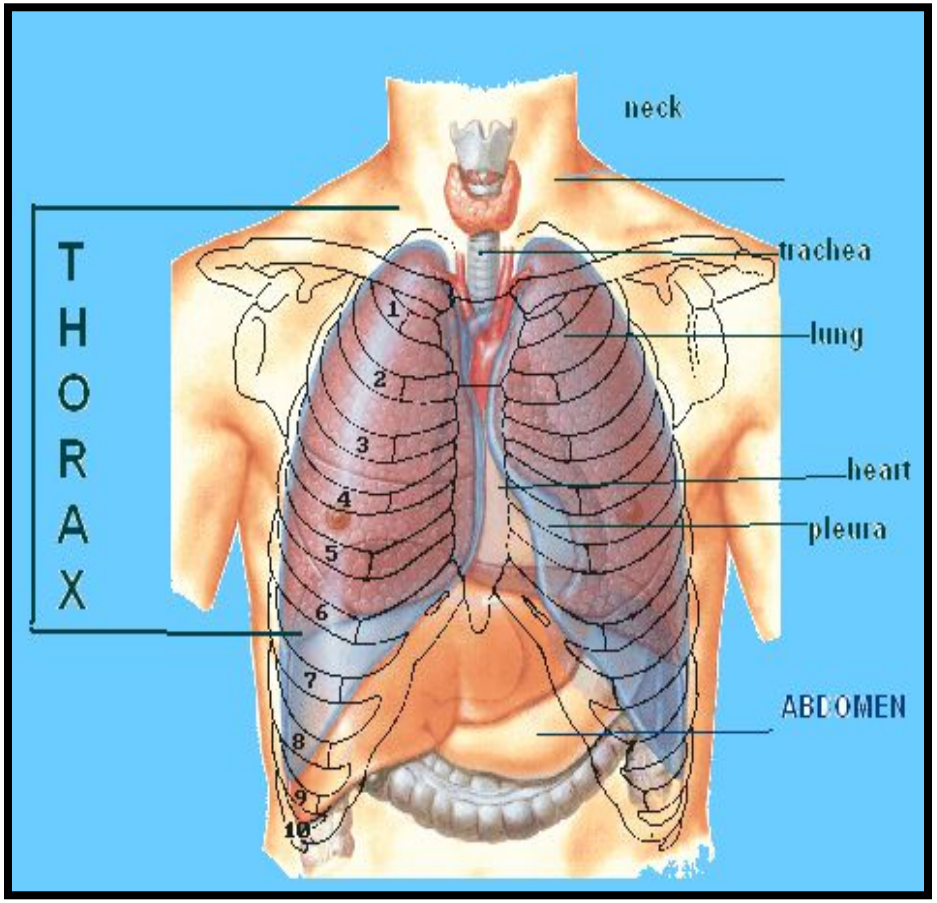
LEARNING OBJECTIVES

At the end of lecture the student should be able to know:

- About structural outline of thorax
- Different components of thoracic cage
- Different compartments of thoracic cavity
- Boundaries of thoracic inlet and outlet

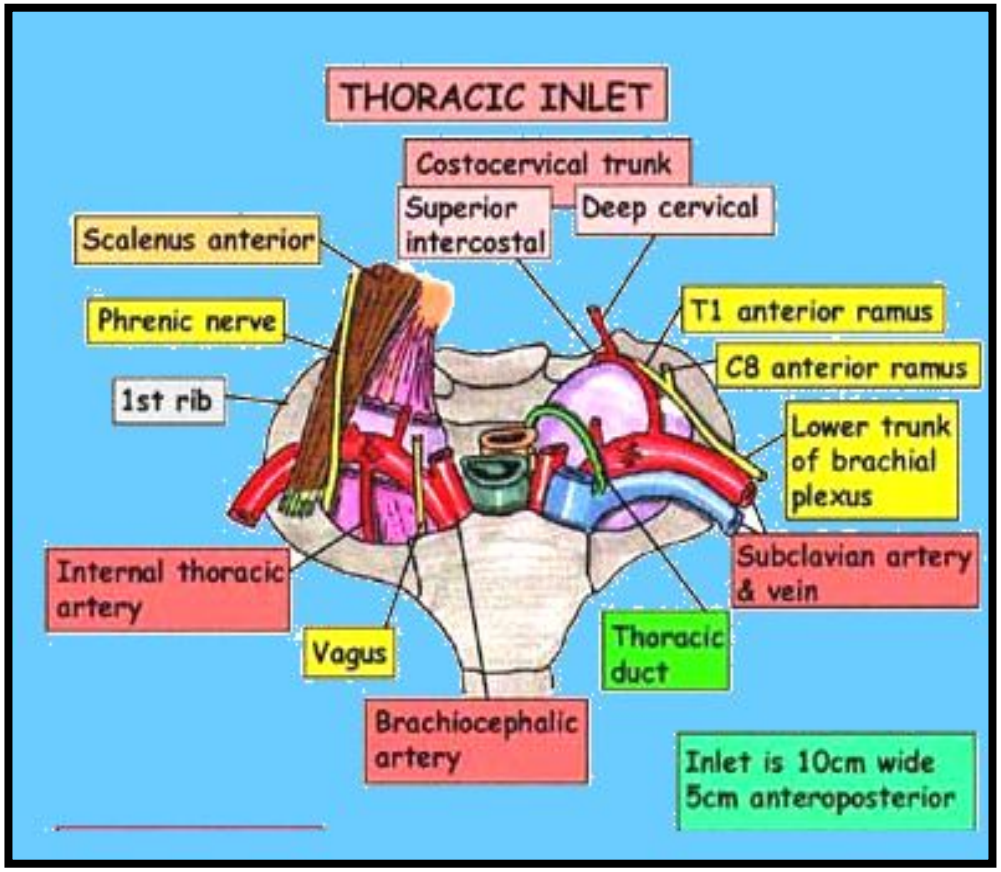
INTRODUCTION

- Region of the trunk between neck and abdomen
- Separated from the abdomen by a partition diaphragm
- Skeletal framework ----thoracic cage
- Cavity contain important cardio respiratory viscera
- Primary function is respiration
- Also helps to protect the viscera important for life



OPENING OF THORAX

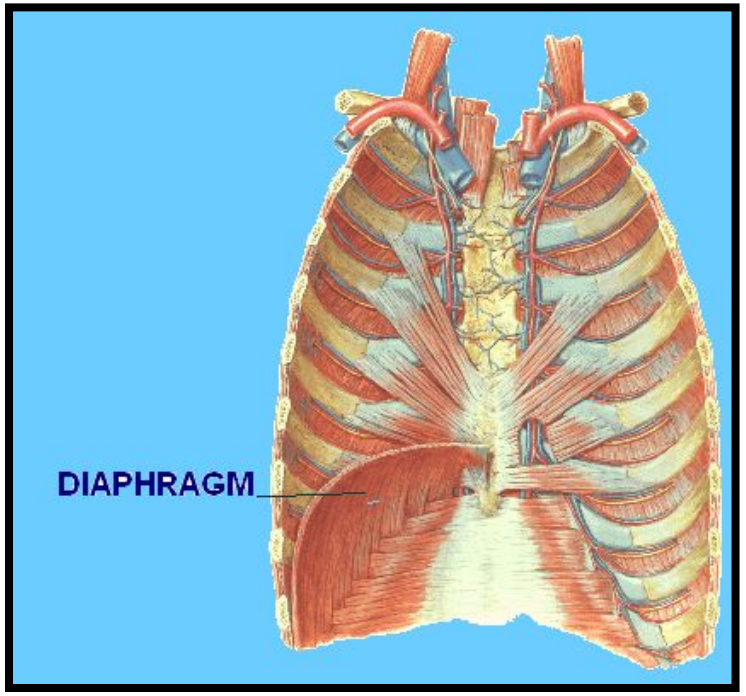
- Communicates with root of neck above through – ***THORACIC INLET*** (known by clinician as outlet)
- Obliquely placed facing downwards and forwards
- Passage for neurovasculature to enter neck and upper limbs

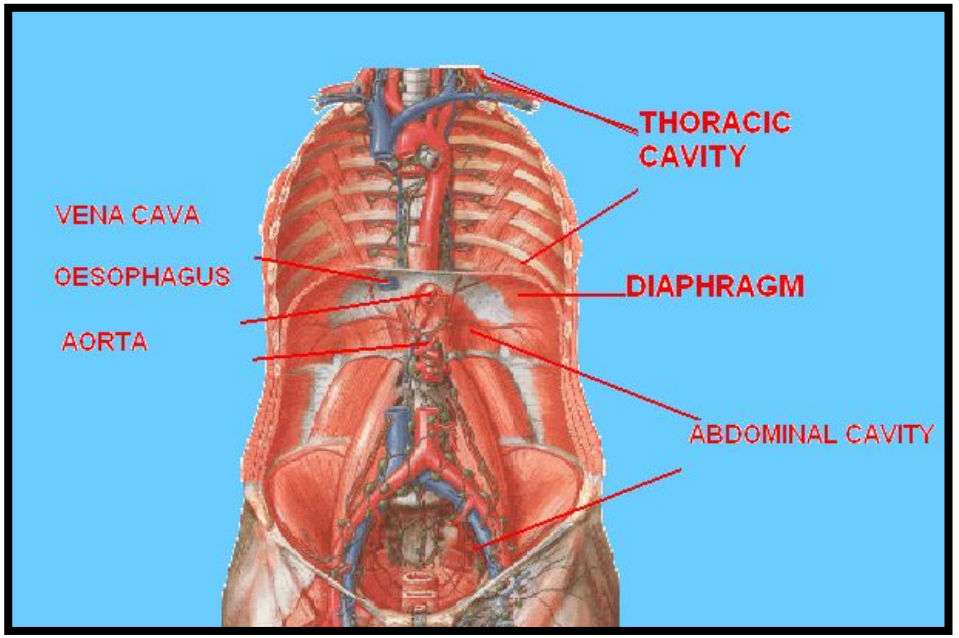


OPENING OF THORAX

Below, communicates with abdominal cavity through large gap; closed by ***musculotendinous partition*** □ ***diaphragm***

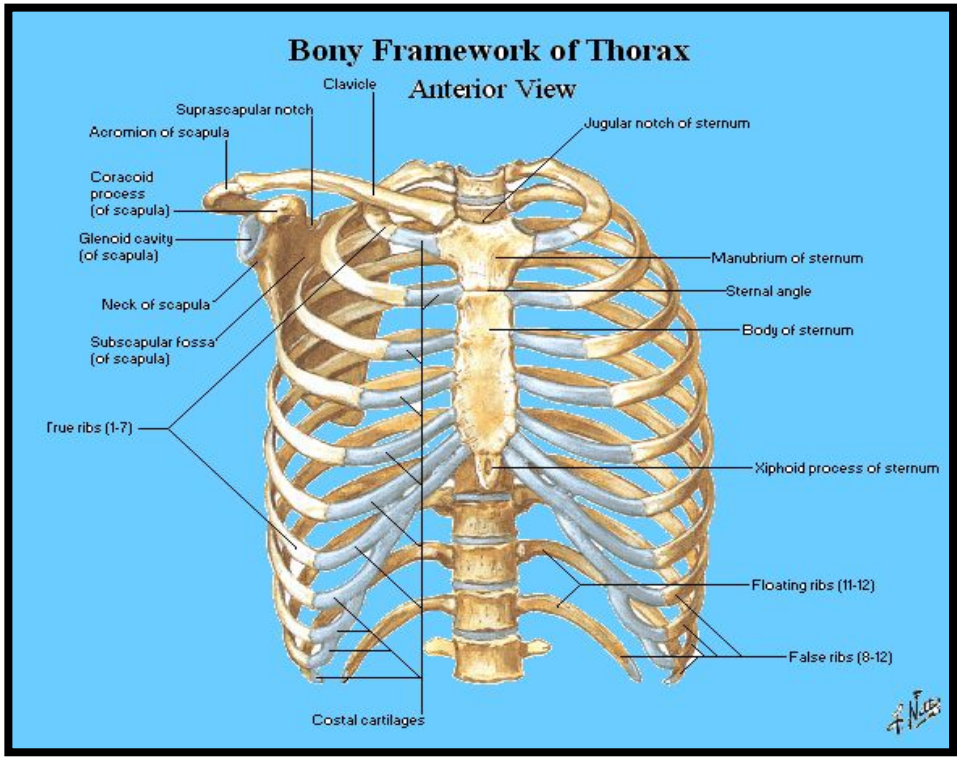
Diaphragm has opening for the oesophagus, aorta and inferior vena cava

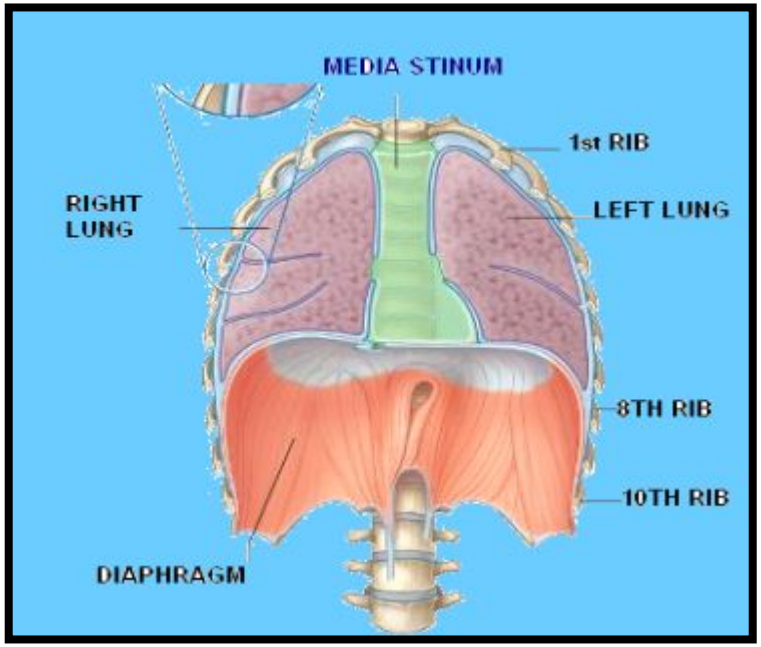




THORACIC REGION

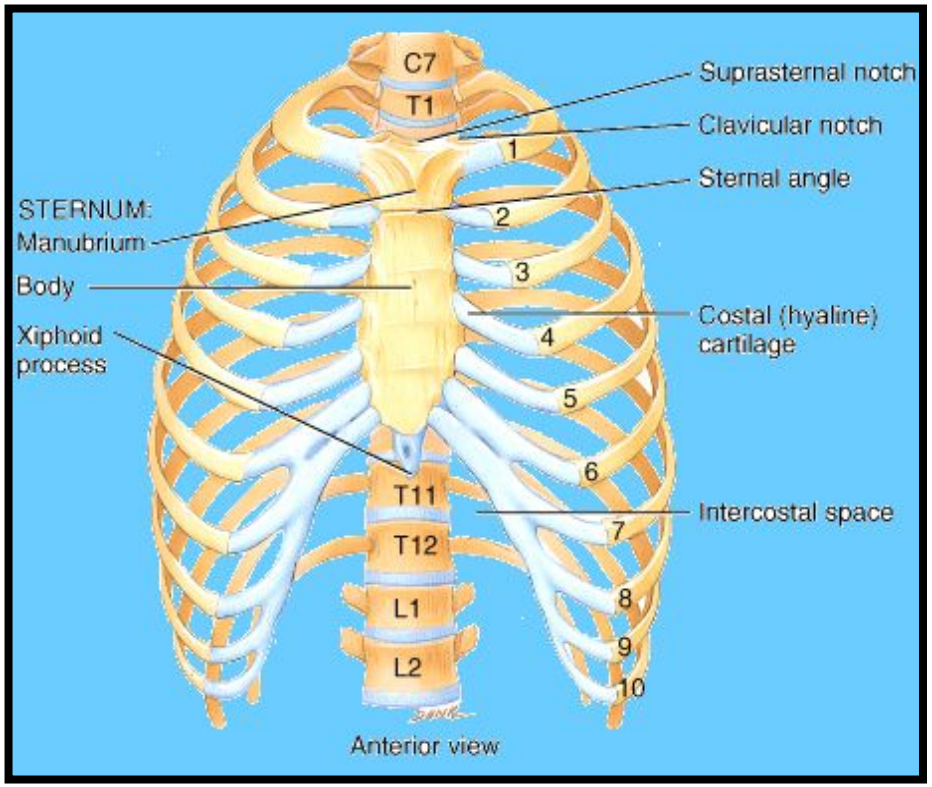
- Thoracic region can be studied in 2 parts
- Thoracic wall
  - Made up of osseocartilaginous framework
- Thoracic cavity
  - Mediastinum
  - Pleural cavity





BOUNDARIES OF THORACIC CAGE

- Anteriorly: by sternum and costal cartilage
- Posteriorly: thoracic part of vertebral column
- Laterally by ribs and intercostal spaces
- Above: suprapleural membrane
- Below diaphragm



STERNUM

Manubrium

- Jugular (sternal) notch
- Articulation with rib #1 & 2
- Clavicular Articular facets
- Sternal Angle – 2nd rib

Body

- Articulates w/ribs 2-7
- Xiphisternal joint

Xiphoid process

- Cartilage-calcifies thru time
- Partial attachment of many muscles

RIBS

Usually, 12 pairs

- 7 **True ribs**-direct attachment to sternum
- 5 **False ribs**-indirect or no attachment to sternum
- **Floating ribs**-make up 2 of 5 False ribs, no ventral attachment

Typical Ribs

- Ribs # 2-9

Atypical Ribs

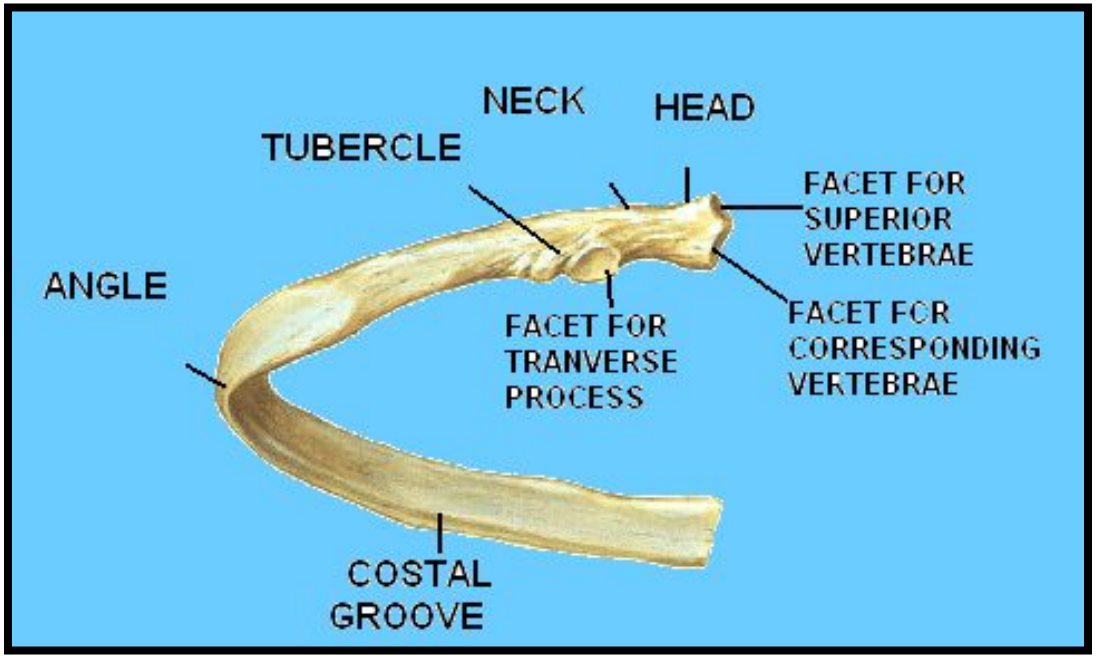
- Ribs #1, 10, 11, 12

Reinforce thoracic cage

RIB ANATOMY

Typical Ribs

- Head
- Neck
- Tubercle
- Angle
- Shaft
- Subcostal Groove

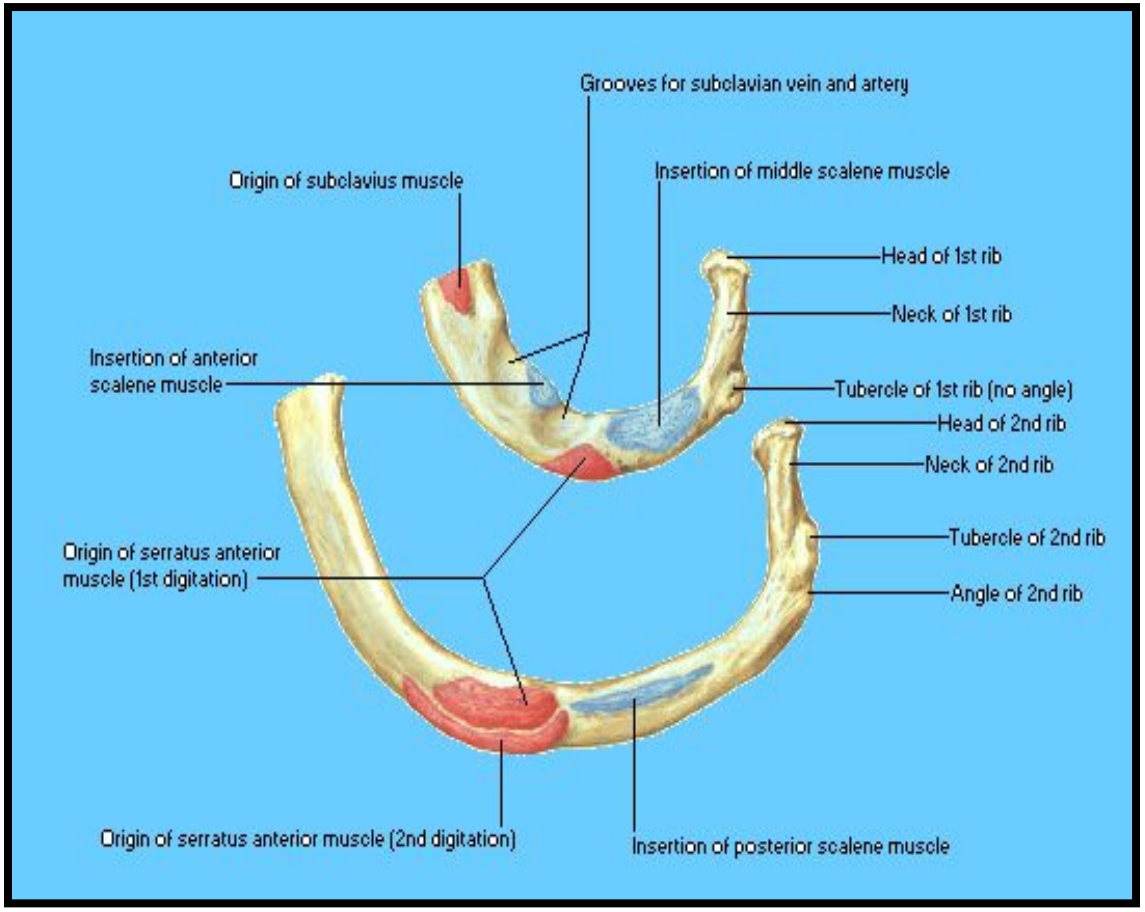


ATYPICAL RIBS

#1-short, flat (S-I), wide, Supports Subclavian vessels

#1, 10-12 articulate with only = # vertebra

#11, 12 don't articulate with transverse processes, or anteriorly at all



RIB ARTICULATION

DORSAL

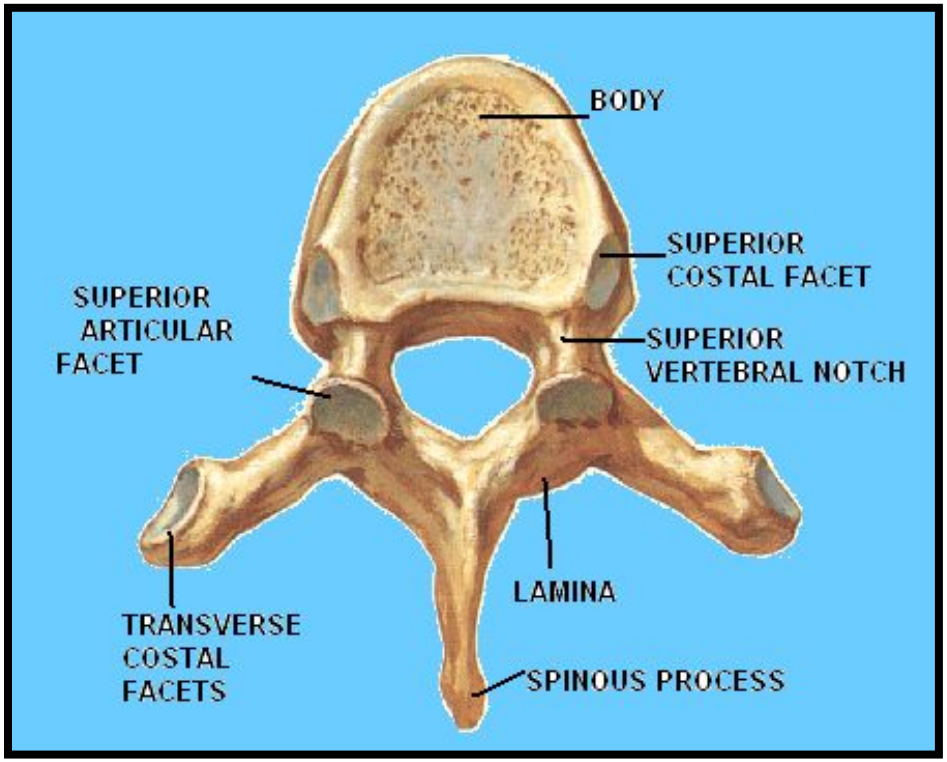
- Tubercle articulates with transverse process
- Head articulates with vertebral bodies

VENTRAL

- Articulates with sternum through costal cartilages

THORACIC VERTEBRAE

- Transverse Costal Facets
- Demifacets on vertebral body
- Spinous Processes long, point inferiorly
- Superior Articular Facets face Dorsally/Posteriorly
- Inferior Articular Facets face Ventrally/Anteriorly
- Vertebral Foramen is Circular
- Body is Heart-shaped



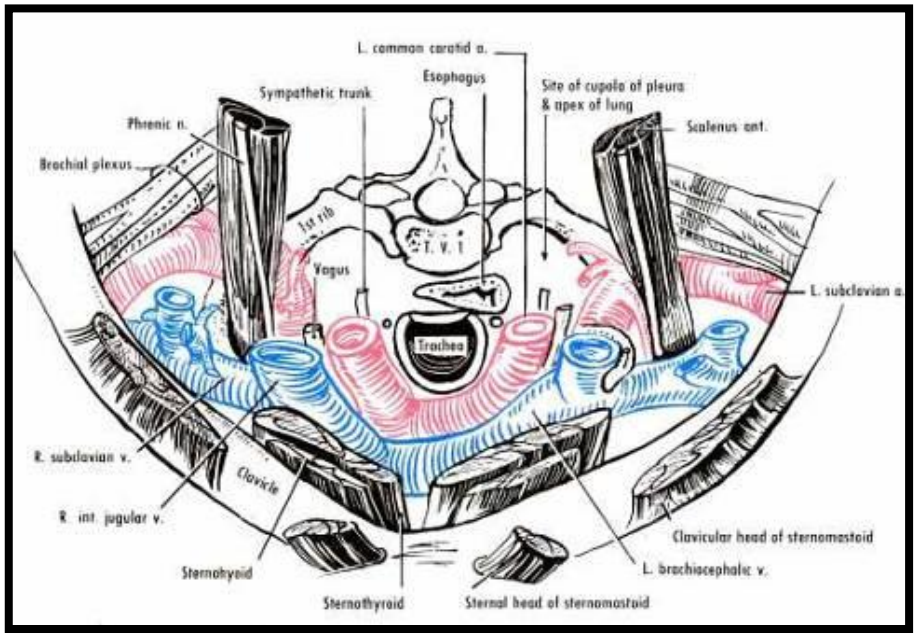
INTERCOSTAL SPACES

- 11 Intercostal spaces
- Bridged by 3 layers of intercostal muscles
- From outside to inside
  - **External intercostal**
  - **Internal intercostal**
  - **Innermost intercostal**
- Neurovascular bundle
  - **Intercostal artery**
  - **Intercostal vein**
  - **Inter costal nerve**
- Run in the lower part of intercostal space near costal groove

THORACIC INLET

Boundaries

- Anteriorly: superior border of manubrium sterni
- Posteriorly : 1st thoracic vertebrae
- Laterally: medial border of 1st rib and their costal cartilages
- Esophagus , trachea and many important nerve and vessels pass through it
- Closed by Sibson's fascia

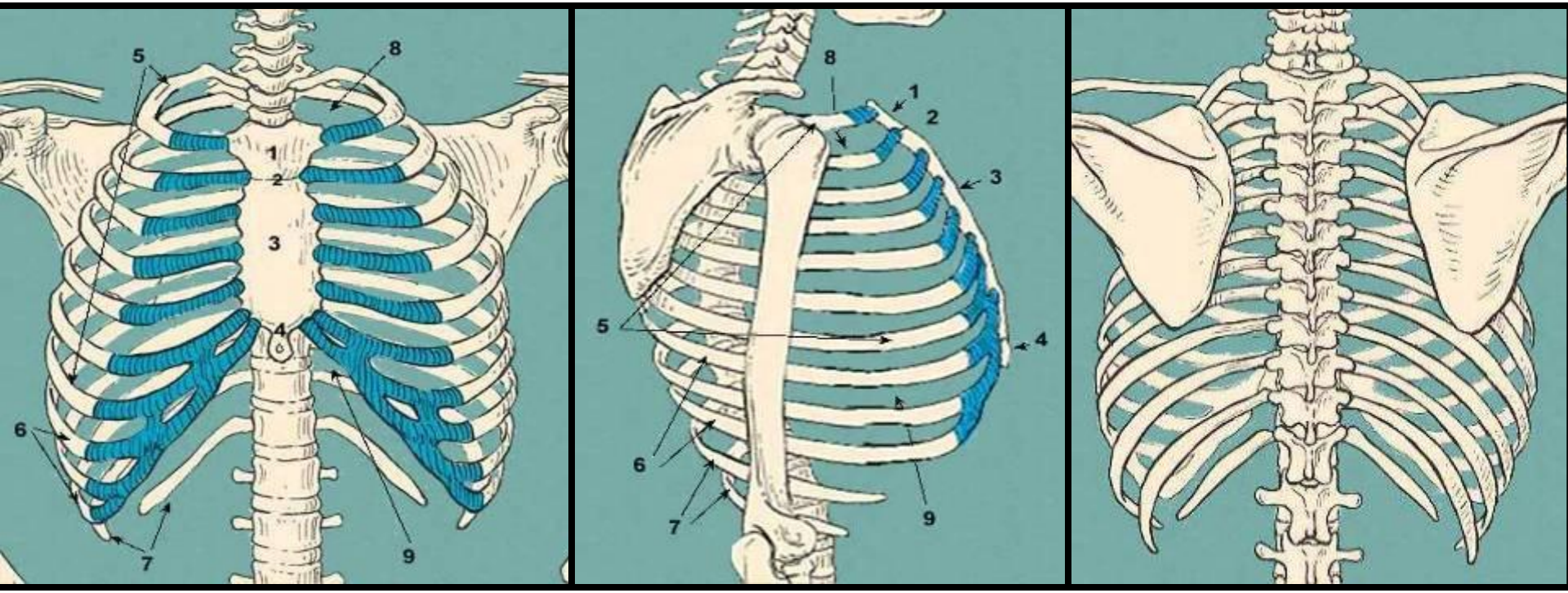


DIAPHRAGMATIC OPENING

- Large opening between thorax and abdomen
- Bounded anteriorly by xiphisternal joint, posteriorly by 12th thoracic vertebrae and laterally by curving costal margin
- Closed by diaphragm

THORACIC WALL

ANTERIOR VIEW LATERAL VIEW POSTERIOR VIEW



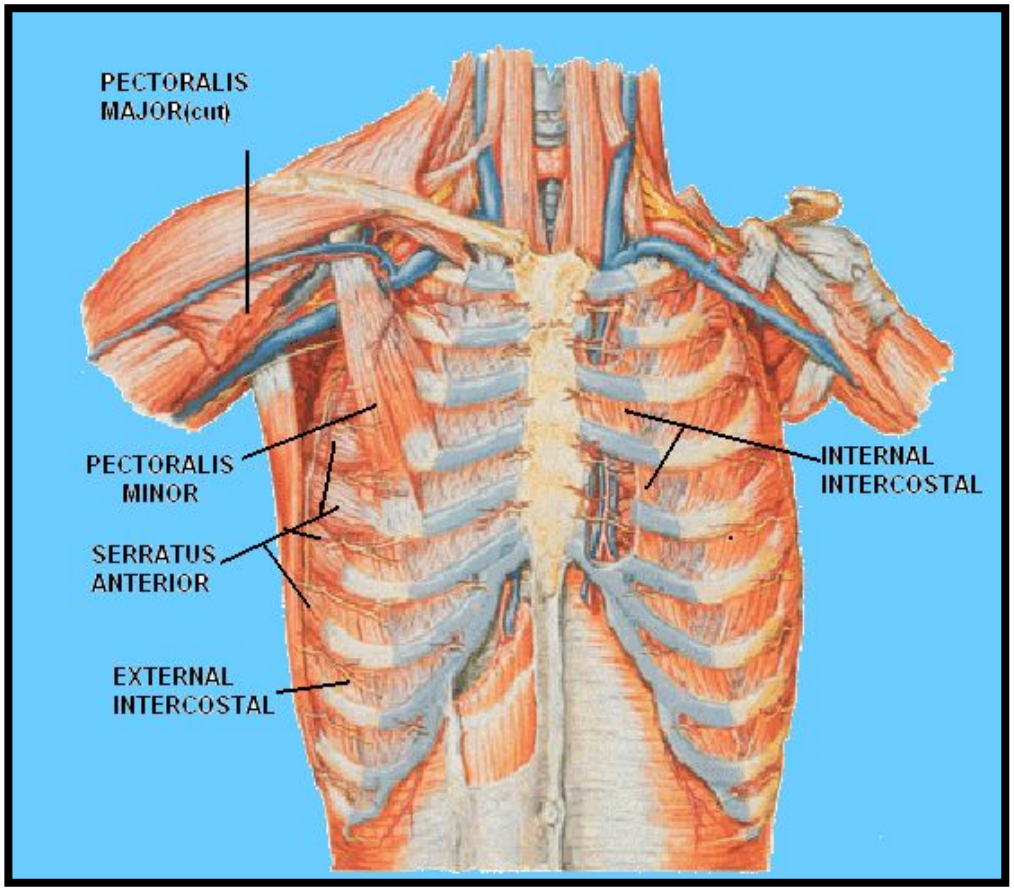
THE MUSCLES OF THORAX

Extrinsic muscles

- Pectoralis major
- Pectoralis minor
- Serratus anterior

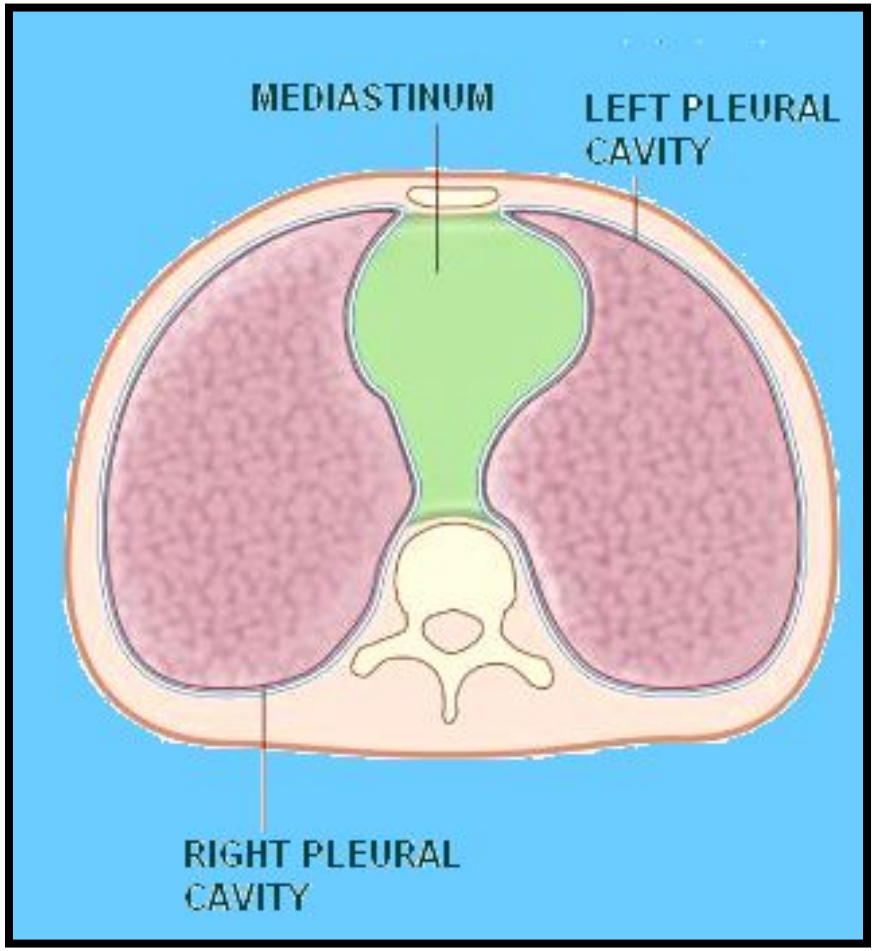
Intrinsic muscles

- Intercostales externi
- Intercostales interni
- Intercostales intimi
- Transverses thoracis



THORACIC CAVITY

- Divided into a central portion mediastinum
- Separates 2 pleural cavities contain lungs

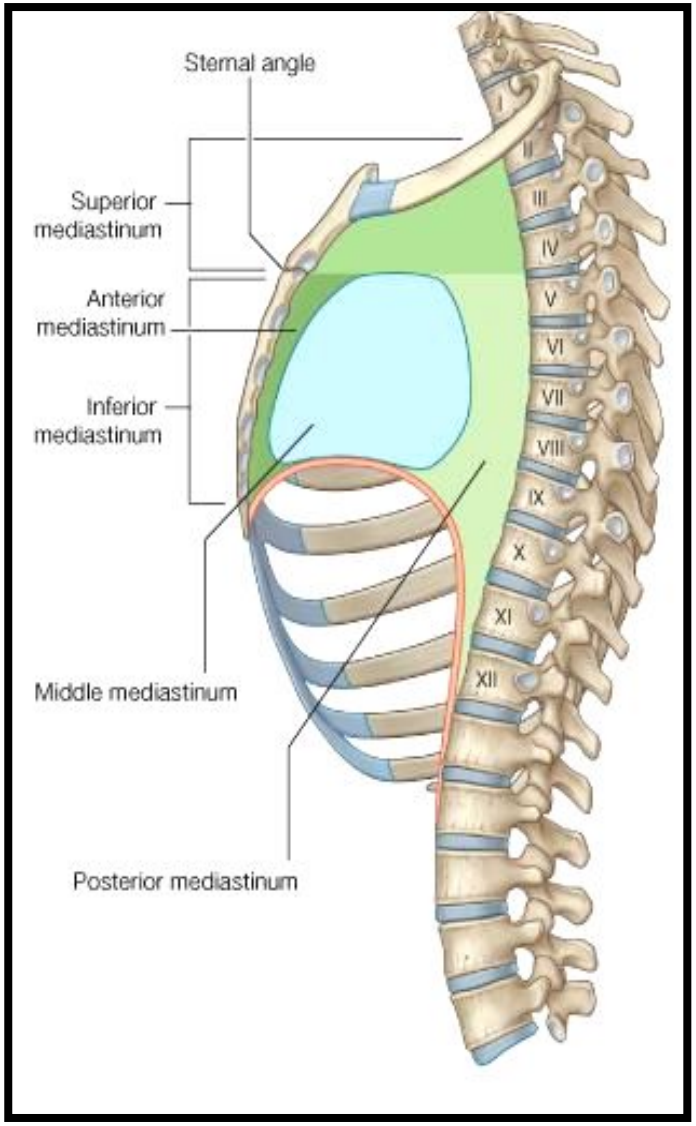


MEDIASTINUM

Is a broad central partition that separates the two laterally placed pleural cavities".

It extends:

- From the sternum to the bodies of the vertebrae; and
- From the superior thoracic aperture to the diaphragm.



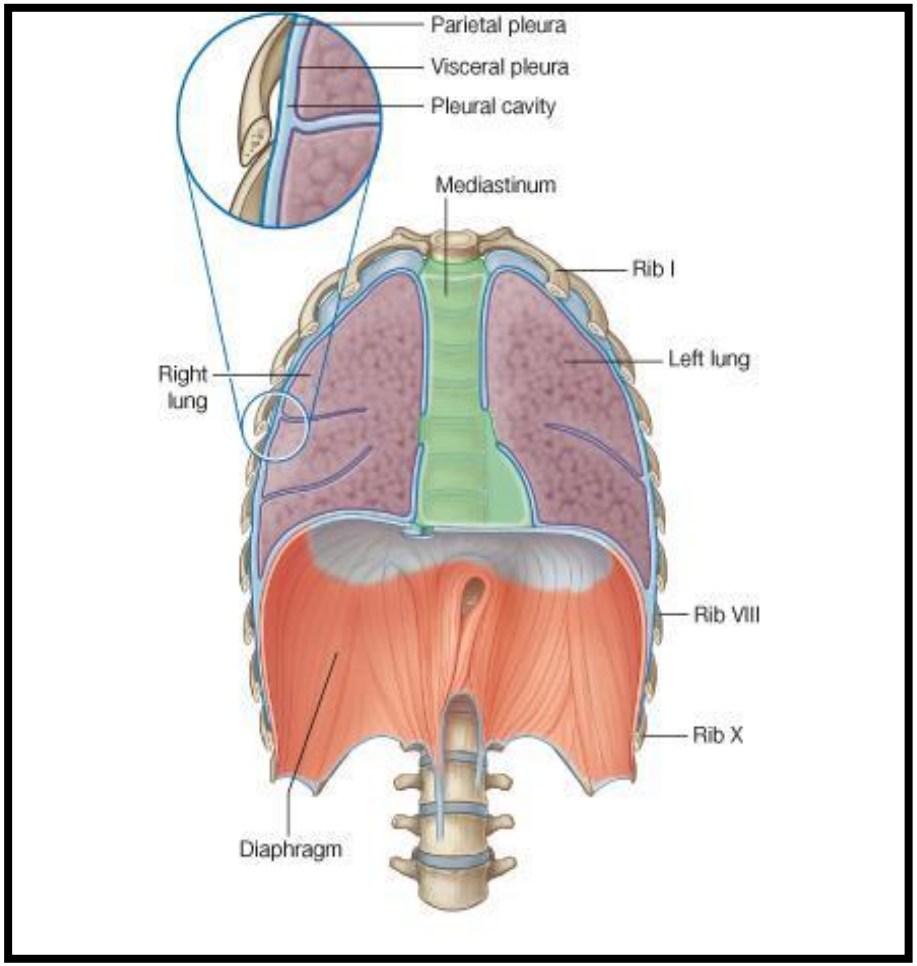
MEDIASTINUM

- Imaginary plane passes through T 4 divides it into superior and inferior mediastinum
- Inferior mediastinum is further divided
- Heart enclosed in pericardium occupies middle mediastinum
- From sternum to anterior pericardium; anterior mediastinum
- From posterior pericardium to vertebrae; posterior mediastinum

PLEURAL CAVITIES

The **pleura** is divided into two major types, based on location:

- Pleura associated with the walls of a pleural cavity is **parietal pleura.**
- Pleura that reflects from the medial wall and onto the surface of the lung is **visceral pleura.**
- Pleural cavity is the potential space enclosed between the visceral and parietal pleurae.



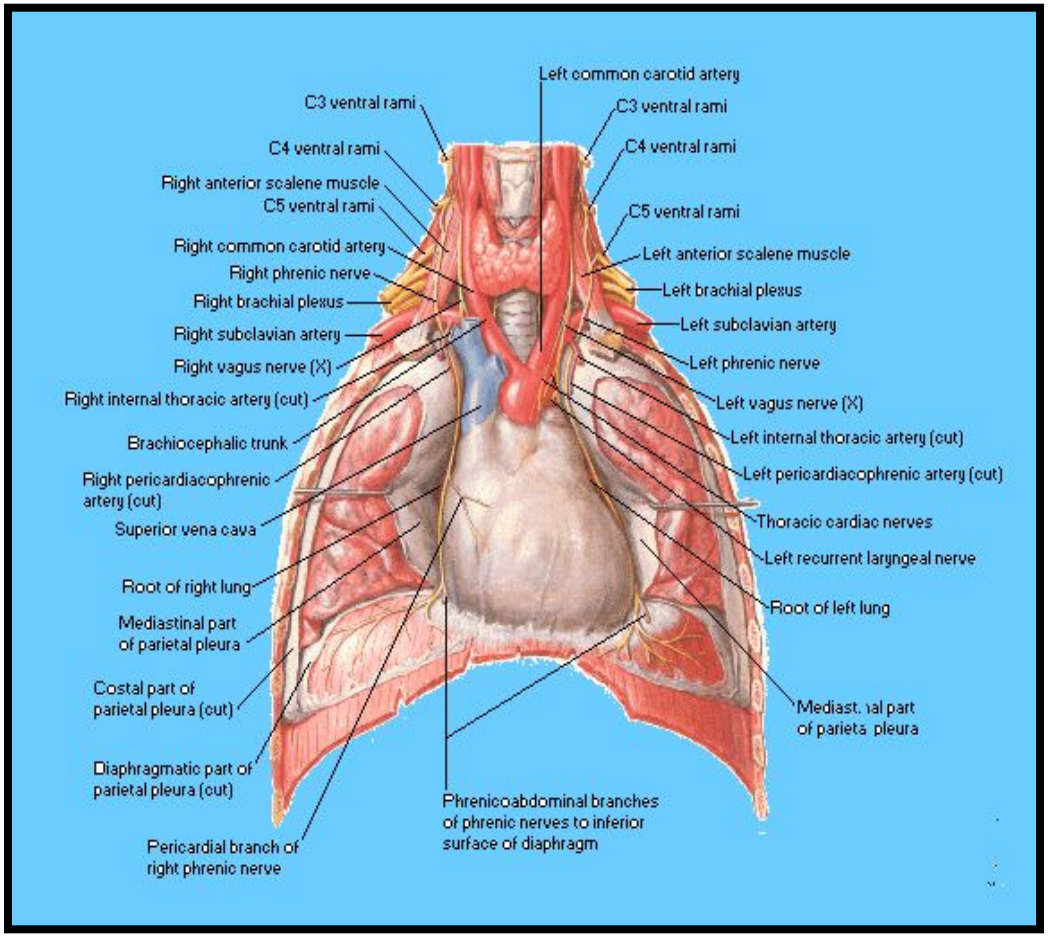
The main thoracic organs which you will examine during study of the thorax are the:

- lungs
- heart

The other structures are:

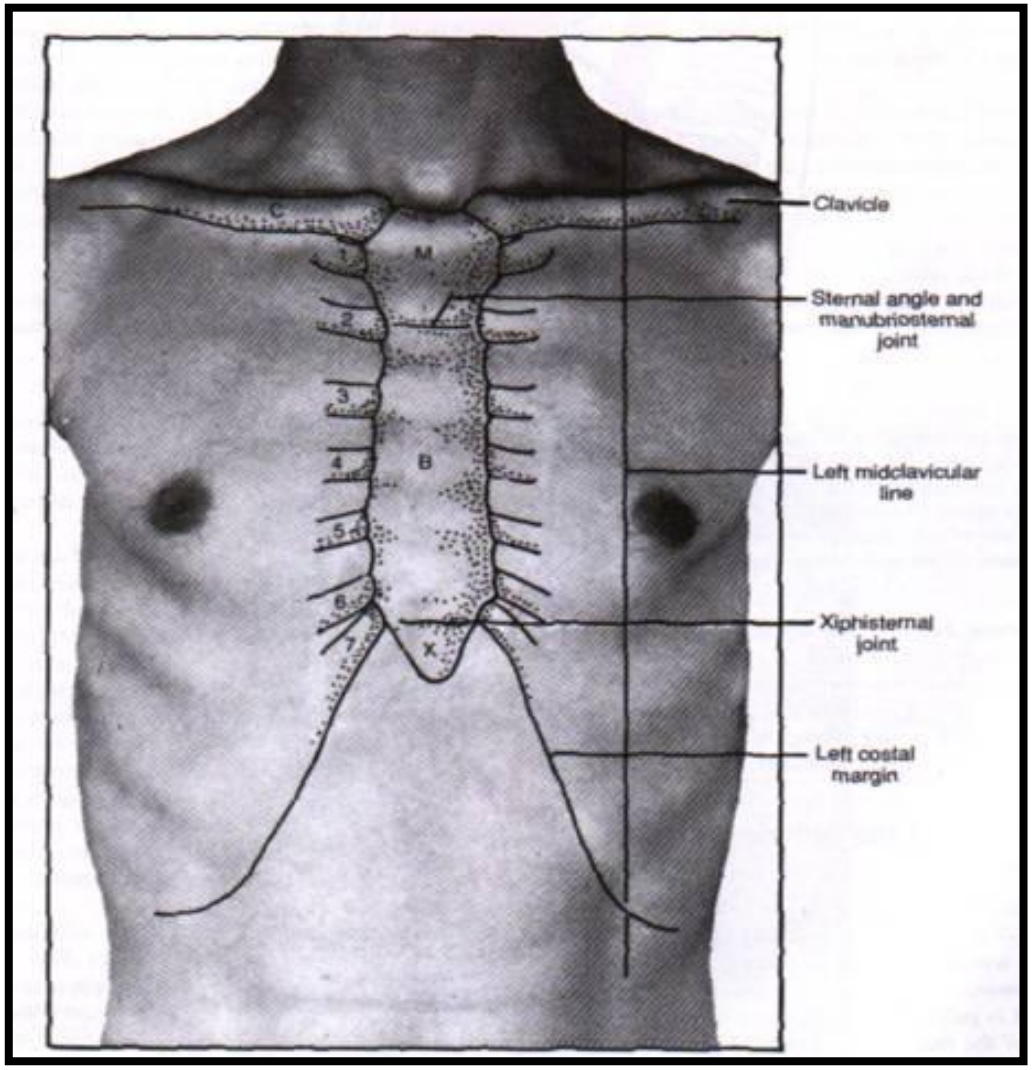
- aorta and its branches
- superior and inferior vena cavae
- trachea and primary bronchi
- sympathetic trunks and their associations
- azygos and hemiazygos venous systems

INTERNAL STRUCTURES OF THORAX



SURFACE ANATOMY

- **ANTERIOR SURFACE**
- Palpate the following
  - Sternum (3 parts)
  - Jugular notch
  - Sternal Angle (= 2nd rib)
  - Clavicle
  - Costal margin
  - Infrasternal angle
  - Xiphisternal joint
- Midclavicular Line
- Midaxillary Line



SURFACE ANATOMY

POSTERIOR SURFACE

- Palpate the following
  - Spinous Process of C7
  - Scapula (ribs 2-7)
    - Scapular spine
    - Acromion Process
    - Inferior Angle of Spine
    - Inferior Border

THANK YOU